Narration Script

LIGTA-S rationale video v2 · how MCS comes into play · version without narration

Length 2:47 · 1920 × 1080 · 30 fps · subtitles burned in · 398 words in four scenes, then a 10-second title card with no narration. Read each line over **LIGTA-S_Rationale_Video_v2_MCS_No_Narration.mp4**. Each line starts when its subtitle appears, and every visual cue is tied to a word, so match your pace to the subtitles.

How to record

1. Play the video and start speaking when each subtitle appears (**Start**).
2. Finish the line by **Finish by**. **Read time** is the time the line needs at an easy pace.
3. Record each scene as its own take. Scene windows: 0:00–0:24.3, 0:24.3–0:58.3, 0:58.3–1:19.8, 1:19.8–2:37.1, then the title card 2:37.1–2:47 (no narration).
4. The boom lands at 2:22.8. Read the conclusion (4.12 to 4.14) after it, then stay silent through the title card to 2:47.
5. Pace is about 2.3 words per second, with a short breath at each comma.

Say it this way

MCS: "M-C-S", letter by letter
LIGTA-S: "LIG-tas", like the Filipino word for safe
PAGASA: pag-A-sa
Signal No. 4: "Signal Number Four"
Reming: REH-ming
Mayon: mah-YON
Albay: al-BYE
lahar: LAH-har
Lola: LOH-lah
"test network", never "Albay", for the E1 numbers
Wind ranges, sources and table values are shown on screen only.

Scene 1 · Storm

0:00.0–0:24.3 · 68 words · bar: Warning

#	Start	Finish by	Read time	Narration (read this) · on screen
1.1	0:00.3	0:04.5	5.2 s	**A typhoon is coming, and PAGASA draws its forecast track toward Albay.** On screen: Map of Bicol at sea level; a typhoon spiral moves in from the Pacific along a dotted forecast track with a widening cone; forecast points +12 h, +24 h, +36 h.
1.2	0:04.8	0:10.5	7.0 s	**Along the track, PAGASA raises wind signals, from Signal No. 1 up to Signal No. 5.** On screen: Signal rings 1 to 5 open around the forecast landfall; the ladder of the five Tropical Cyclone Wind Signals builds on the right (wind range per signal).
1.3	0:10.8	0:18.3	9.6 s	**Each signal comes with a lead time: thirty-six hours at Signal No. 1, but only twelve at Signal No. 4 or 5.** On screen: The lead-time column lights up: 36 h, 24 h, 18 h, 12 h, 12 h. Source: PAGASA Tropical Cyclone Wind Signals (2022).

1.4	0:18.6	0:23.6	7.8 s	**Families must leave before the winds arrive. But where do they go, and is there room for Lola?** On screen: A family of four by their house; bubbles "Where do we go?" and "Room for Lola?".

Scene 2 · Community

0:24.3–0:58.3 · 73 words · bar: Learning

#	Start	Finish by	Read time	Narration (read this) · on screen
2.1	0:24.7	0:26.5	2.6 s	**Albay has lived through this before.** On screen: Albay as a field of 720 barangay dots; Mayon at the centre. PSA 2020 census.
2.2	0:26.8	0:34.1	7.0 s	**In 2006, Typhoon Reming sent lahar down Mayon, and about 1,200 people were dead or missing.** On screen: Card: "November 2006 · Typhoon Reming (Durian) · lahar from Mayon's old deposits · about 1,200 dead and missing". Source: Philippine Daily Inquirer, 2016.
2.3	0:34.4	0:40.2	5.7 s	**In 2020, Super Typhoon Rolly made landfall in Albay under Signal No. 5.** On screen: Card: "November 2020 · Super Typhoon Rolly (Goni) · landfall in Tiwi, Albay · Signal No. 5". Source: PAGASA, 2020.
2.4	0:40.5	0:51.5	9.1 s	**And when Mayon rose to Alert Level 3 in January 2026, four schools became evacuation centers, and 3,424 learners were affected.** On screen: Card: "January 2026 · 4 schools became evacuation centers · 3,424 learners affected". Source: Philippine Daily Inquirer, 2026.
2.5	0:51.8	0:57.5	7.4 s	**The law makes schools a last resort. Goals 4, 11, and 13 meet in the same classroom.** On screen: Chips SDG 4.a, SDG 11.5 and SDG 13.1 meet over one classroom; "RA 10821: schools are evacuation centers only as a last resort".

Scene 3 · Stakes

0:58.3–1:19.8 · 61 words · bar: Time

#	Start	Finish by	Read time	Narration (read this) · on screen
3.1	0:58.6	1:03.4	7.0 s	**As the track shifts, the plan must shift too: which zones leave, and which roads close.** On screen: Badge "SIGNAL NO. 4"; a lead-time clock counts down from 12 h; test network E1 with its hazard zones.
3.2	1:03.7	1:07.6	5.2 s	**Heavy rain sends lahar down the river channels and closes the bridges.** On screen: Lahar channels A and B cut across the road ring; bridges A1 to A4 and B1 to B3 turn red with a cross.

3.3	1:07.9	1:12.9	6.5 s	**Send everyone to the nearest site, and families arrive at shelters that are already full.** On screen: Arrows to the nearest shelters; counter "898 arrive at full shelters" (test network E1, Alert Level 4 with typhoon lahar).
3.4	1:13.2	1:19.0	7.8 s	**So how do we shelter everyone before the lead time runs out, the vulnerable first, and classrooms last?** On screen: "Everyone sheltered · Vulnerable first · Classrooms last · Before the lead time ends" with a question mark.

Scene 4 · How MCS comes into play

1:19.8–2:37.1 · 196 words · bar: MCS

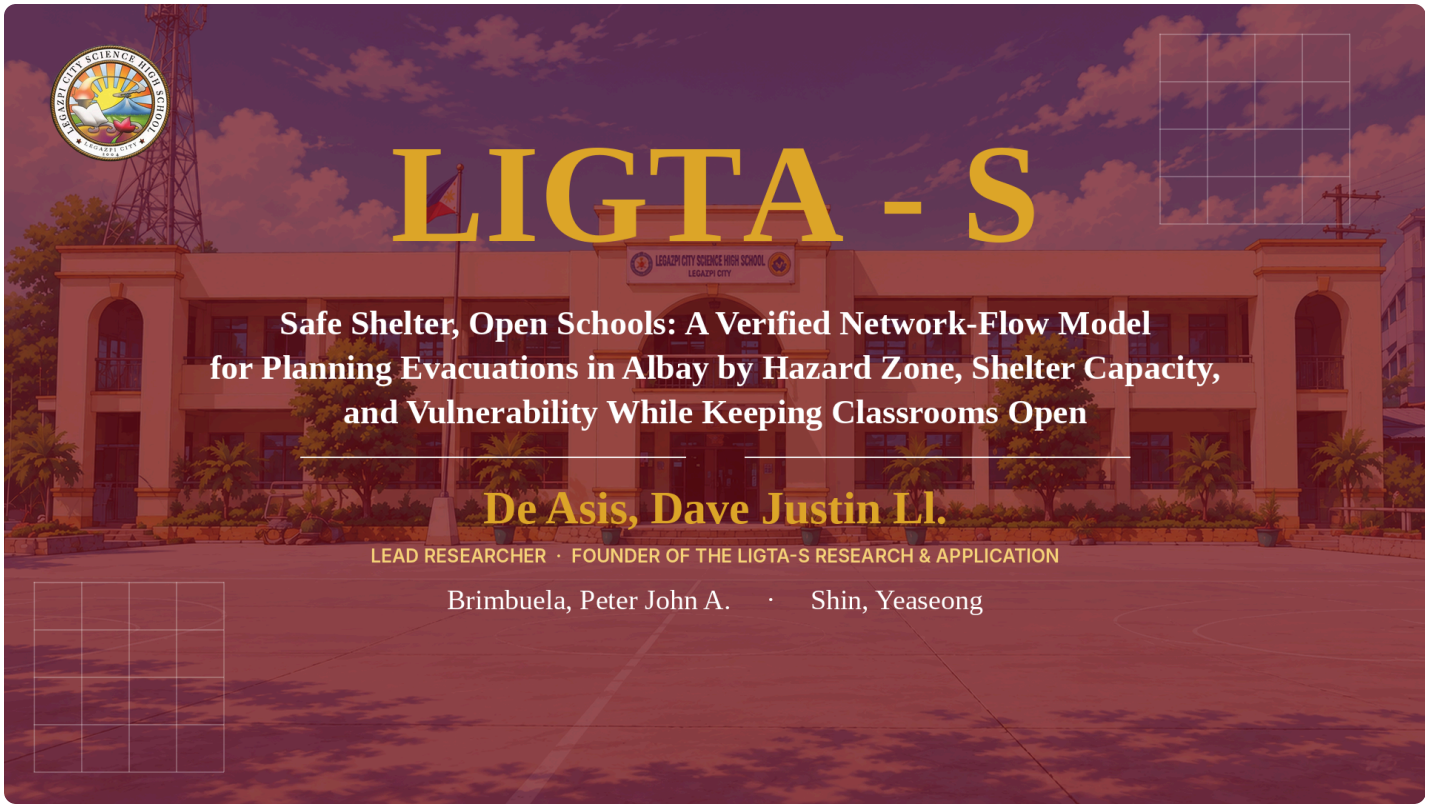
#	Start	Finish by	Read time	Narration (read this) · on screen
4.1	1:20.1	1:22.2	3.0 s	**This is where MCS comes into play.** On screen: The letters M, C and S drop in and unfold into Mathematical, Computational, Sciences.
4.2	1:22.5	1:30.4	10.0 s	**Objective one: turn the signal and the forecast track into a hazard scenario, the zones that must leave and the roads that close.** On screen: Panel OBJECTIVE 1 · SCENARIO: σ = (Z_σ, X_σ); signal + track → zones (lahar, flood, storm surge) and closed roads; test network E1 with closed bridges.
4.3	1:30.7	1:38.9	7.4 s	**Objective two: write the law's order as one lexicographic minimum-cost flow: vulnerable first, everyone next, classrooms last.** On screen: Panel OBJECTIVE 2 · SHELTER PLAN: three shelter tiers, the lexicographic objective and its penalties; evacuee flows animate.
4.4	1:39.2	1:46.4	8.3 s	**Objective three: compute the exact clearance time on a time-expanded network, and check it against the signal's lead time.** On screen: Panel OBJECTIVE 3 · CLEARANCE TIME: T* = min{T : F(T) = D}; T* ≤ lead time; arrival curve F(T).
4.5	1:46.7	1:54.1	8.3 s	**Objective four: find the bridges to protect before landfall, and site new Ligtas Pinoy Centers with a proven guarantee.** On screen: Panel OBJECTIVE 4 · BRIDGES AND CENTERS: Δ_B = T*(σ + B) − T*(σ) ≥ 0; f(S_g) ≥ (1 − 1/e) f(S*); the new center lights up.
4.6	1:54.4	1:59.2	4.8 s	**Objective five: prove every result, then verify it against independent code.** On screen: Panel OBJECTIVE 5 · PROOF: ten proof marks turn into checks.
4.7	1:59.5	2:06.7	7.8 s	**On the test network, losing one lahar channel's bridges stretches the clearance time from 53 to 95 minutes.** On screen: Card 1: "Clearance time 53 → 95 min · lahar channel A closed · protect it first" (test network E1, synthetic).
4.8	2:07.0	2:13.9	8.3 s	**With typhoon rain on an eruption, 450 people are cut off, and one new center reaches 400 of them.** On screen: Card 2: "450 cut off → one new Ligtas Pinoy Center reaches 400".

4.9	2:14.2	2:18.5	5.2 s	**At Alert Level 3, the plan closes fourteen classrooms instead of forty.** On screen: Card 3: "Classrooms closed 40 → 14 · learners displaced 1,702 → 532".
4.10	2:18.8	2:22.4	3.5 s	**Ten results, proven, then verified against independent code.** On screen: Ten green checks; 300/300 plans, 660/660 networks, 4,796 checks, 0 violations.
4.11	2:23.0	2:24.9	2.6 s	**LIGTA-S will not stop a typhoon.** On screen: Boom. Screen darkens; the line appears in white.
4.12	2:25.2	2:31.1	8.3 s	**But before landfall, it will tell every family where to go, in time, and keep the most classrooms open.** On screen: The line appears in gold.
4.13	2:31.4	2:33.2	3.5 s	**A signal tells you the storm is coming.** On screen: "A signal tells you the storm is coming." in white.
4.14	2:33.5	2:35.7	4.3 s	**A plan tells you if there is room, in time.** On screen: "A plan tells you if there is room, in time." in gold.

Title card · Researchers

2:37.1–2:47 · no narration

#	Start	Finish by	Read time	Narration (read this) · on screen
5.1	2:37.1	2:47	—	**[No narration. Stay silent; let the music and the title card play.]** On screen: the school building under a deep red overlay, the school seal top-left and a white corner grid; the gold letters LIGTA - S drop in one by one with a hit, then a gold shine; the full paper title in white; two thin white lines draw outward; "De Asis, Dave Justin Ll." in large gold with "LEAD RESEARCHER · FOUNDER OF THE LIGTA-S RESEARCH & APPLICATION"; then "Brimbuela, Peter John A. · Shin, Yeaseong" in white. Fade to black at 2:47.



Title card, final frame (also supplied as LIGTA-S_Title_Page_1920×1080.png and LIGTA-S_Title_Page_3840×2160.png).

Clean read

The same narration without the production notes. Use it as a teleprompter or to rehearse. A slash marks a short breath.

SCENE 1 · STORM · 0:00.0–0:24.3

0:00.3 A typhoon is coming, / and PAGASA draws its forecast track toward Albay.

0:04.8 Along the track, / PAGASA raises wind signals, / from Signal No. 1 up to Signal No. 5.

0:10.8 Each signal comes with a lead time: thirty-six hours at Signal No. 1, / but only twelve at Signal No. 4 or 5.

0:18.6 Families must leave before the winds arrive. But where do they go, / and is there room for Lola?

SCENE 2 · COMMUNITY · 0:24.3–0:58.3

0:24.7 Albay has lived through this before.

0:26.8 In 2006, / Typhoon Reming sent lahar down Mayon, / and about 1,200 people were dead or missing.

0:34.4 In 2020, / Super Typhoon Rolly made landfall in Albay under Signal No. 5.

0:40.5 And when Mayon rose to Alert Level 3 in January 2026, / four schools became evacuation centers, / and 3,424 learners were affected.

0:51.8 The law makes schools a last resort. Goals 4, / 11, / and 13 meet in the same classroom.

SCENE 3 · STAKES · 0:58.3–1:19.8

0:58.6 As the track shifts, / the plan must shift too: which zones leave, / and which roads close.

1:03.7 Heavy rain sends lahar down the river channels and closes the bridges.

1:07.9 Send everyone to the nearest site, / and families arrive at shelters that are already full.

1:13.2 So how do we shelter everyone before the lead time runs out, / the vulnerable first, / and classrooms last?

SCENE 4 · HOW MCS COMES INTO PLAY · 1:19.8–2:37.1

1:20.1 This is where MCS comes into play.

1:22.5 Objective one: turn the signal and the forecast track into a hazard scenario, / the zones that must leave and the roads that close.

1:30.7 Objective two: write the law's order as one lexicographic minimum-cost flow: vulnerable first, / everyone next, / classrooms last.

1:39.2 Objective three: compute the exact clearance time on a time-expanded network, / and check it against the signal's lead time.

1:46.7 Objective four: find the bridges to protect before landfall, / and site new Ligtas Pinoy Centers with a proven guarantee.

1:54.4 Objective five: prove every result, / then verify it against independent code.

1:59.5 On the test network, / losing one lahar channel's bridges stretches the clearance time from 53 to 95 minutes.

2:07.0 With typhoon rain on an eruption, / 450 people are cut off, / and one new center reaches 400 of them.

2:14.2 At Alert Level 3, / the plan closes fourteen classrooms instead of forty.

2:18.8 Ten results, / proven, / then verified against independent code.

2:23.0 LIGTA-S will not stop a typhoon.

2:25.2 But before landfall, / it will tell every family where to go, / in time, / and keep the most classrooms open.

2:31.4 A signal tells you the storm is coming.

2:33.5 A plan tells you if there is room, / in time.

TITLE CARD · RESEARCHERS · 2:37.1–2:47

[Silent. Title card: LIGTA-S · De Asis, Dave Justin Ll., Lead Researcher · Founder · Brimbuela, Peter John A. · Shin, Yeaseong. Ends 2:47.]